Machine Learning for Computational Science and Engineering (2026) 2:16
https://doi.org/10.1007/s44379-026-00065-z

RESEARCH

Early diagnosis of cardiovascular disease using stacking ensemble augmented by optimized machine learning framework

Divya Mishra[1] · **Amit Agrawal**[2] · **Ankush Mittal**[1]

Received: 16 September 2025 / Accepted: 9 April 2026 / Published online: 16 April 2026
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2026

Abstract

Cardiovascular disease (CVD) is still a major cause of death globally, and hence the need for a precise and robust predictive model for early detection. While many machine learning models have been developed, most of these are based on single classifiers, simple validation techniques, or inadequate robustness analysis. To overcome these challenges, this paper presents an optimized stacking ensemble model for early cardiovascular disease prediction based on structured clinical data. The proposed model combines three diverse base models, namely Random Forest, XGBoost, and CatBoost, with a meta-model of Logistic Regression for enhanced predictive robustness and generalization. The proposed model was tested on the Heart Statlog Hungary Cleveland dataset, which has 1190 patient samples and 14 clinical features. A structured data preprocessing technique was designed, which included missing data treatment, outlier identification, and feature scaling to cope with the heterogeneity of the clinical data. Hyperparameters of individual base models were tuned using grid search. For testing the robustness of the model, the framework was tested on various train and test splits (70:30, 80:20, and 90:10) as well as cross-validation. The stacking model designed in this study performed better, with a maximum accuracy of 97.98%, precision of 98.57%, recall of 95.83%, F1-score of 97.18%, and AUC-ROC of 98.64%. The above results show that the optimized stacking ensemble model has better predictive reliability and fewer false negatives than the individual models. This study provides a reliable and practical decision-support system for the early diagnosis of cardiovascular disease.

Keywords Machine learning · Catboost · Xgboost · Ensemble learning · Cardiovascular disease

1 Introduction

Cardiovascular disease (CVD) ranked consistently as the leading cause of pragmatic mortality globally [1]. According to World Health Organization (WHO) reports, CVD and stroke cause 17.5 million deaths globally each year. The majority of the more than 75% of heart disease-related deaths take place in middle- and low-income nations. Furthermore, 80% of all demises from CVDs [2] are attributable to heart attacks and strokes. Cardio disease is frequently diagnosed after a physical examination and observation of the patient's symptoms. Smoking, aging, having a history of heart disease in family, increase in blood pressure, obesity, diabetes, stress, high cholesterol, and inactivity are risk determinants for cardiovascular disease [3]. Some of these risk factors may be lowered by changing one's lifestyle to include things like exercising, reducing stress, stopping smoking, and losing weight. To detect heart issues, typically a patient's past medical records, medical imaging and physical examination tests i.e., blood tests, cardiac MRIs, electrocardiograms, and echocardiograms are used.

Medication, changes in lifestyle, bypass surgery of coronary artery, angioplasty, and implanted devices such as defibrillators or pacemakers can all be used to treat cardiovascular disease [4]. With the help of the huge volumes of patient data that are readily available due to the expansion of modern healthcare systems, it is now feasible to create models for predicting cardiovascular disease. Large datasets are analyzed from multiple perspectives using machine learning,

✉ Divya Mishra
divya19aug@gmail.com

Amit Agrawal
amitagrawal1909@gmail.com

Ankush Mittal
dr.ankush.mittal@gmail.com

[1] Computer Science and Engineering, COER University, 7 Km Roorkee Haridwar Highway, Roorkee 247667, Uttarakhand, India

[2] Wells Fargo, Bangalore 610101, Karnataka, India

which is regarded as a data-sorting technique that turns the findings into concrete knowledge [5].

Machine Learning (ML) techniques like as Support Vector Machine (SVM), Artificial Neural Network (ANN), Decision Tree (DT), Logistic Regression (LR), and Random Forest (RF) have been used in a number of research to evaluate medical data and forecast heart conditions. In a recent study, [6] employed machine learning models to forecast a multiethnic population's likelihood of developing heart disease. To stratify CVD risks, the scientists combined a sizable dataset of electronic health record data with sociodemographic data. The multiethnic population's CVD risk was accurately predicted by the models. Likewise, a deep learning (DL) system was used in a different study by [7] to forecast coronary artery disease (CAD). To train the DL model, the researchers used coronary computed tomography angiography (CCTA) pictures and clinical data.

Several machine learning techniques were utilized in a study by [8] used to forecast CVD using clinical data. The researchers used models like DT, K-nearest-neighbor (KNN), and RF. The authors found that these models give accurate result in CVD prediction. Similarly, a study by [9] used ML methods to determine the heart disease risk factors. The researchers investigated the information from National Health and Nutrition Examination Survey (NHANES) as well to determine coronary heart disease risk factors.

In another study [10] the author looked at how well different ML algorithms predicted heart conditions. According to the authors, the models were highly accurate in forecasting heart conditions. The ML classifiers employed in this study have demonstrated encouraging outcomes in terms of identifying CVD [11–23] risk. High accuracy has been demonstrated by the RF, LR and KNN algorithms in identifying CVD risk. When compared to single classifiers, ensemble learning strategies like bagging, voting and Adaboost have increased classification accuracy. By enabling early detection using advanced ML method like ensemble learning along with individualized therapy, it may enhance patient outcomes.

Artificial Intelligence (AI) has a number of applications in the medical field, it may assist medical practitioners in predicting CVD, for example, based on clinical features, namely age, sex of subjects, type of chest pain, fasting blood sugar, no. of main vessels, blood pressure: resting, ST slope, Cholesterol, Max heart rate, Thalassemia, Resting ECG, OLD Peak, Exercise-induced angina, among others [14]. Various studies have been conducted, which give different methods utilized by medical practitioners in predicting CVD in medical institutions in their early stages; which has been explained in Section II. Both ML & Deep Learning (DL)-based models provide a basis for proposing intelligent models. The models have an innate connection whereby DL consists of models from machine learning, which in turn consists of models from AI. There exist a number of models under machine learning, which include SVM, RF, KNN, Naive Bayes (NB), LR, & XGBoost Classifier. Models under deep learning include RNN, CNN, & deep neural networks (DNN) [7].

The need for addressing the limitations of existing methods for predicting cardio disease and exploring the possibility of using ML techniques for improved diagnosis accuracy is what has primarily driven this work. In fact, it is one of the most precise combined machine learning algorithms with minimal complexity of computation. For the Hungary-Cleveland HD data set, testing the proposed hypothesis has been employed. Grid Search was the optimization environment used in this work for hyperparameter learning of the intelligent model.

This intelligent system's performance was measured on different train-test ratios of dataset i.e., 70:30, 80:20, and 90:10. A benchmark study was also carried out to see how well it performed in comparison to previous studies. The suggested framework offers a viable way to diagnose HD accurately and quickly, with possible advantages for the treatment and decision-making of cardiac patients.

1.1 Contributions of work

This paper doesn't propose a new base learning algorithm but instead offers a framework-level contribution to clinical decision support. The main contributions of this work are:

(i) a clinically driven stacking ensemble method combining Random Forest, XGBoost, and CatBoost algorithms with Logistic Regression to produce fewer false negatives in CVD predictions;
(ii) robustness evaluation with multiple train and test splits (70:30, 80:20, 90:10) as against single split evaluation;
(iii) an analysis of clinical errors using a confusion matrix that emphasizes trade-offs between recall and precision that are important in initial screening;
(iv) Comparative evaluation that shows steady improvement in recall and AUC-ROC values over best boosting benchmarks.

This study is organized in the sections that follows: The CVD related work is discussed in Sect. 2, Sect. 3 describes the problem statement and Sect. 4 discusses material and methodology, Sect. 5 discusses data collection, Sect. 6 experimental results followed by conclusion in Sect. 6 & future work in Sect. 7.

2 Related work

Narayan et al. [15] proposes machine learning techniques, including SMOTE oversampling, that can be used to diagnose cardiac disease with best accuracy. Ramanathan [14] employed different number of Machine learning techniques, especially SVM that can predict risk of cardio disease accurately using clinical-based data. Sofany et al. [16] proposed a method for early prediction of cardiac disease using feature selection techniques like chi-square, ANOVA, and mutual information.

After testing ten machine learning algorithms on both public and private datasets, they determined that XGBoost was the best classifier. SHAP was employed for model interpretability, and SMOTE was used to address data imbalance. They have created a smartphone software that predicts cardiac problems in real time.

An approach using the Swarm-ANN model was adopted by S. Nandy et al. [17]. The DT, KNN, SVM, NB, LR, and RF classification algorithms were some of the popular classification algorithms utilized to evaluate the Cleveland dataset with 14 attributes. For the approach using the Swarm-ANN model, the authors achieved an accuracy of 0.9578.

N. Long et al. [18] introduced a method that merges a fuzzy logic system based on interval types with a rough set-based attribute reduction. The Firefly Algorithm has been utilized to perform the feature reduction process. The authors evaluated performance indicators for both the heart disease (HD) and Single Proton Emission Computed Tomography (SPECTF) datasets in order to determine the efficacy of their methodology.

The authors report that 94.2% sensitivity, 87.2% accuracy, and 68.9% specificity were attained by the SPECTF dataset. On the other hand, the HD dataset showed a 93.3% specificity, 88.3% accuracy, and 84.9% sensitivity. The authors clocked accuracy of 92.84% on a real-time dataset and 94.78% on the Cleveland dataset using 6-Fold Cross-Validation (CV). Furthermore, a number of models were proposed by S. Ayon et al. [19] and evaluated using two-fold CV (5 and 10). These models included LR, DT, DNN, NB, RF, and KNN.

Recent works have attempted to investigate new modalities in detecting cardiovascular disease. As an example, a hybrid model based on transformer encoders and residual convolutional networks was developed in order to recognize cardiovascular disease from heart sound signals and verify the efficacy of deep learning on heart sound signals. In addition, such methods require complex data acquisition techniques and heavy computations, thus making it difficult to implement in clinical settings [16].

Srinivas et al. [20] introduces a hyper parameter-tuned XGBoost Classifier called hyOPTXhg for predicting cardio disease using the Optuna framework. Their strategy used Min–Max normalization for data preparation in conjunction with the Optuna optimization technique. A maximum accuracy of 94.71% was attained by the model when it was tested on different datasets, Cleveland (303 samples, 14 features) and Heart Failure dataset from Kaggle (299 samples, 13 features.

To enhance HD prediction, Vote with Logistic Regression and Naïve Bayes classifier were used as a hybrid model by Amin et al. [21]. The dataset that was used in the research was Cleveland HD dataset, who combined feature selection with the suggested model. The result confirms that vote model gives accuracy of 87.31% with nine important features. For HD classification, Mohan et al. [22]. A hybrid RF based linear model (HRFLM) was used to perform the test on Cleveland. The results define the best accuracy of 88.4% followed by 0.82,0.92,0.90,0.90 as specificity, sensitivity, precision and f1 score. Using the Cleveland dataset. In their research, the work of Javeed et al. [23] introduced the ensemble model of intelligent system for HD prediction named Random Search Algorithm with RF (RSA-RF), which was compared with other ML classifiers such as SVM, RF, Adaboost, extra trees (ET). In their work, the authors were able to achieve a accuracy of 93.33% for their RSA-RF with the accuracy rate of the proposed model in the research work being performed via the different train-test based validation strategy. For HD diagnosis based on the Cleveland data set, the work of Budholiya, K., and others. [24] introduced the diagnosis intelligence technique of XGBoost optimized for HD prediction using 14 features with the precision of 91.80%.

The optimized stacked SVM intelligent method for the early prediction of HF was introduced in the work of Ali et al. [25]. A holdout validation approach was used by the authors to assess the efficiency of the model. Using eight hyperparameters, this model claims an accuracy of 92.22% using the Cleveland dataset. Ali et al. [26] proposes a diagnostic intelligent method that works automatically for the prediction of HD and works on the $\chi 2$ Statistical methodology, which is well integrated with DNN. In order to determine the model's efficiency and the achieved accuracy of 93.3%, the authors employed a holdout validation technique and the Cleveland dataset for HD prediction. Dileep et al. [27] created sovereign model for the early prediction of CVD, using Cleveland dataset along with HD dataset, and employing the cluster-based bi-directional long-short term memory (C-BiLSTM) technique. They reported 94.39%, 92.41%, 92.76%, 91.18%, and 94.21% accuracy for the DNN, LR, DT, NB and KNN models using 5-Fold CV respectively.

J. Azmi et al. [28] suggested a thorough literature evaluation of HD detection using several machine learning algorithms. The authors intended to classify the latest ML-based

approaches utilized in HD diagnosis using the unbalanced dataset.

G.N. Ahmad et al. [29] looked into several ML-based methods, i.e., LR, SVM, KNN and XGB, for HD diagnosis, both with and without the GridSearchCV optimization methodology. M. M. Ali et al. [30] demonstrated a comparative analysis of a number of supervised ML algorithms, i.e., RF, KNN, AdaboostM1, DT and Multilayer Perceptron (MLP), using the feature significance approach for HD prediction. They then compared performance metrics with those of other approaches.

For imputation of null values, they utilized the multivariate imputation by chained equations (MICE) approach. Five strategies—MMC, Adaptive, Random, QUIRE, and AUDI with/without hyperparameter tuning (HT) were utilized to develop a multi-label active learning-based model for the prediction of heart disease in the work done by El-Hasnony et al. [31]. They tuned the hyper parameters using a grid search-based optimization method, and they examined the performance using f1 Score and accuracy. M. G. El-Shafiey et al. [32] proposed a hybrid RF method that blends Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) in HD prediction task. They used the 10-Fold & Holdout processes for cross validation, comparing them to both conventional and well-established approaches. According to Deepika D. et al. [33], an efficient HD classification method was provided using a MLP for improved Brownian Motion-based on the Dragonfly Algorithm (MLP-EBMDA). They looked at a few selected metrics, such as kappa, accuracy, recall, precision, and f1 Score, in order to validate the presented approach. H. Hasanova et al. [34] presented a Sine Cosine Weighted K-Nearest Neighbor (SCA-WKNN) method for HD prediction, which learns from blockchain based data. The following metrics were used to compare the SCA-WKNN system to other systems: accuracy, computation time, recall, precision, root mean square error and f1 Score. K. Budholiya et al. [24] proposed an improved diagnostic intelligence technique based on XgBoost for early detection of HD. They used various metrics i.e., accuracy, specificity, area under the curve, sensitivity, and f1 Score, to assess the effectiveness of the proposed model. Additionally, they used the Bayesian optimization technique to adjust hyperparameters.

Recent studies use AI as a tool for Valvular Heart Disease (VHD) diagnosis from Phonocardiography (PCG) signals to avoid instances of misdiagnosis. Models based on CNNs, such as the VGG16 and ResNet50 models, are common, and Transformer models handle the temporal relationships between cardiac signals. The experiment protocols include testing models on standard benchmarks such as PhysioNet 2016 and Yaseen, with multi-class and binary classification. Normalization of MFCC, DCT, and corruption or data augmentation are common pre-processing methods. The combined CNN-Transformer models perform better than standalone models, with accuracy and AUC scores well over 97% [35].

Recently, a study integrated Bald Eagle Search Optimization (BESO) into several machine learning classifiers for CAD prediction, demonstrating its performance using two heterogeneous datasets. In this view, advanced feature selection with extensive preprocessing and a comparison with traditional clinical risk scores were considered a remarkable strength, especially toward the enhancement of predictive accuracy with reduced feature redundancy. However, reliance on a single 70–30 holdout validation strategy without cross-validation or statistical significance testing may limit confidence in generalizability. While much emphasis has been placed on feature optimization in this study, ensemble learning strategies or robustness analysis across different splits of data were not pursued, thus leaving scope for further improvement in model stability and clinical reliability [36].

However, most of these works report results based on single train-test splits, mostly without detailed error analysis or focusing on recall, though it is crucial in medical screening. On the other hand, very few studies have focused on the robustness across varying data availability scenarios. These observations bring us to the focus of the current study, which places more emphasis on recall-oriented evaluation and robustness analysis using multiple data splits.

Recently, a deep learning model for the prediction of cardiovascular disease was developed using Binary Krill Herd optimization for feature selection and a capsule network classifier. The model showed better predictive accuracy on the Cleveland dataset than conventional machine learning techniques, emphasizing the potential of meta-heuristic optimization and deep learning models for risk prediction in clinical settings [37].

In the current research, a stacked ensemble model for the prediction of cardiovascular disease was proposed using a neural network reasoning model and Hawks Optimizer. The model overcame the problem of class imbalance and showed better predictive accuracy than conventional models on a benchmark dataset [38].

In this research [39], the researchers proposed an ensemble-based cardiovascular disease prediction model using k-Nearest Neighbor, XGBoost, AdaBoost, and Random Subspace classifiers with feature selection based on Linear Support Vector Feature Measure. The proposed model was tested on the UCI dataset and showed better prediction performance and decision support compared to the existing methods.

In this work [40], the authors proposed a hybrid deep learning architecture (DenseNet-ABiLSTM) for multiclass arrhythmia classification based on PPG signals. The proposed model effectively integrated convolutional feature

extraction, sequence modeling, and attention-based learning to obtain excellent performance in distinguishing multiple arrhythmias, thus proving the efficacy of deep learning for automatic cardiac rhythm classification.

In this work [41], various linear and nonlinear dimensionality reduction algorithms were assessed for the classification of cardiac arrhythmias from ECG signals. Following wavelet-based feature extraction, unsupervised dimensionality reduction was performed before classification with a Probabilistic Neural Network. The findings indicated that nonlinear algorithms, such as fastICA, provided better classification results than the conventional PCA.

2.1 Critical analysis and research gaps

Despite the encouraging results from existing literature on the application of machine learning and ensemble models for cardiovascular disease forecasting, there are several limitations that need to be addressed. Most existing literature on this topic uses single-validation methodologies, such as fixed splitting patterns for training and testing, which may result in biased model accuracy and negatively impact overall generalizability. Moreover, few existing solutions have tried to optimize for accuracy alone without much consideration for other aspects like robustness, statistical validation, or stability for varying distributions. Although bagging, boosting, voting, and similar ensemble models have shown promising improvement over basic classifiers, few models have employed the need for comprehensive optimization or the selection of meta-learned models. Moreover, the models are not very explainable, and most existing solutions have been tested on considerably smaller datasets from single sources, thereby limiting their practical applicability. Most recent literature shows a growing trend towards optimized ensemble models, comprehensive validation methodologies, and well-rounded performance metrics; still, there is relatively untapped territory related to optimized stacking methodologies that are comprehensive in their assessment and concurrently stable in their generalization performances.

Table 1 provides a summary of these investigations, and Fig. 1 depicts the entire process flow of our proposed system. The dataset goes through preprocessing and feature engineering. This is followed by model building using the Random Forest, XGBoost, and CatBoost algorithms. The hyperparameters are tuned separately for each of the base learners. The best of the predictions are finally aggregated using Logistic Regression as the meta-learner.

3 Problem statement

A dataset $D = \left\{ \left(x_i, y_i \right) \right\}_{i=1}^{m}$, where:

- $x_i \in R^n$ is the feature vector of n clinical parameters.
- $y_i \in \{0,1\}$ is the binary label (0 : NO CVD, 1 : CVD)
- The dataset is break down into training and test ratios:

$$D_{\text{train}} \cup D_{\text{test}} = D, \left| \mathrm{D}_{\text{train}} \right| = r \cdot m \tag{1}$$

Where $r \in \{0.7,0.8,0.9\}$ is the training ratio.

4 Materials and methodology

The process flow for the classification of cardiovascular diseases has been shown in Fig. 1. The approach employed here starts with the preprocessing of the data to improve the quality of the data. The preprocessed dataset will then be split into various split sizes (70:30, 80:20, and 90:10) to evaluate the influence of the split sizes on the testing results. Finally, the performance assessment of the machine learning classifier will be done after employing the best possible hyperparameters.

4.1 Proposed stacking ensemble framework

The proposed framework adopts a structured multi-stage pipeline approach for the prediction of cardiovascular disease. The overall approach is designed to ensure the quality of the data, optimization of the model, and effective prediction results. The overall workflow is explained below.

- Step 1: Data Acquisition

The clinical dataset was acquired from the Heart Statlog Hungary Cleveland dataset, which includes the demographic and medical variables of the patients related to the diagnosis of cardiovascular disease.

- Step 2: Data Preprocessing

The raw dataset was preprocessed to enhance the quality of the data. Missing values in the dataset were addressed, and outliers in the dataset were identified using statistical range analysis. The dataset was normalized to ensure uniform scaling of the variables. This step helps to eliminate noise in the data and facilitate faster convergence of the model.

- Step 3: Feature Preparation

All the clinical variables were formatted into structured feature vectors, which represent each patient record. The target variable was specified as a binary variable related to the presence of cardiovascular disease.

Table 1 Literature review of Cleveland Heart disease dataset

Author	Year	Methodology	Findings	Limitations
Narayan et al. [15]	2024	Supervised ML classifiers with SMOTE oversampling technique	95.45 (90:10-train test ratio)	Although high accuracy is achieved using SMOTE oversampling, the experiment on imbalanced datasets depends on a single 90:10 train/test split, which could result in overly optimistic outcomes and inadequate evaluation of generalization performance
Ramanathan [14]	2024	SVM	91.51	The proposed approach for SVM has good predictability, but it only focuses on a single classifier and lacks research on ensemble learning, robustness assessment, and validation on more than one division of data
Sofany et al. [16]	2024	XAI	97.5	Even though it is an effective integration of EKI methods, there is little consideration for robustness analysis, cross-validation, and methods to prevent overfitting, making it difficult to gauge the stability and practicability of the results in the clinical environment
S. Nandy [17]	2021	Swarm-ANN	95	Its accuracy is high but involves a large amount of computational complexity and is tested on a small number of data examples, with less emphasis on testing scalability and robustness and using more than one validation approach
Long et al. [18]	2015	RF	84.9	Although innovative when proposed, in the study, the approach is actually tested on older datasets with less variation in features, and it does not compare with the performance of modern ensemble or optimization-driven machine learning methods
S. Ayon et al. [19]	2020	Various ML techniques	94.28 of KNN	Within the study, the comparative analysis involves a variety of classifiers, but the validation strategy is less superficial, with less focus on statistical testing, prevention of overfitting, and stability of performances on varying distributions of data
Srinivas et al. [20]	2022	hyOPTXg	94.71	The Optuna-optimized model XGBoost improves the accuracy of the model, but the paper does a limited interpretability analysis and lacks careful robustness or generalization testing across diverse datasets and several validation conFigureurations; therefore, its results are not valid

Table 1 (continued)

Author	Year	Methodology	Findings	Limitations
Amin et al. [21]	2019	NB and LR	87.41	As indicated in the research, while feature selection enhances the predictive accuracy, the performance of the hybrid voting model appears to be moderate and sensitive to the specific features of the data, thus challenging the prospects of scalability in a diverse clinical setting
Mohan et al. [22]	2019	HRFLM	88.4	In the context of the research endeavor undertaken by this study, the hybrid RF linear method improves the effectiveness of classification, but the lack of significance testing and cross-validation trails raises questions about the resilience to overfitting
Javeed et al. [23]	2019	RSA-RF	93.33	In the proposed RSA-RF model, the accuracy reported is high using the holdout validation method, though the fact that cross-validation and robustness were not performed is questionable, especially with regards to its level of sensitivity to the division of the data set at hand
Budholiya et al. [24]	2020	Optimized XGBoost with One-HOT encoding	91.80	Although the Bayesian optimization enhances the performance of the XGBoost algorithm, the method presented in the research paper relies greatly on the feature encoding methods, instead of exploring diversity or resiliency on ensembles when it comes to the validation approaches
Ali et al. [25]	2019	L1/L2 SVM (Linear), Grid Search	92.22	The optimal stacked SVM system has better accuracy in classification, though its validation only occurs in one manner, so it does not provide enough evidence regarding its quality
Ali et al. [26]	2019	$\chi 2$ model with DNN	93.3	The χ^2-integrated deep learning model is competitive in terms of accuracy but not interpretable and only validated on a single data set, which may hinder its acceptance in the clinical field as described in the research study
Dileep et al. [27]	2022	C -BiLSTM	94.78	The predictive model developed using the C-BiLSTM performs well; though, complexity may act as a scaling factor for the possibility of implementing it in real time in clinical decision support systems, as discussed in the study
Azmi et al. [28]	2022	SVM, DT, NB, RF, KNN, ANN, LR,	The authors intended to classify the latest ML-based approaches utilized in HD diagnosis using the unbalanced dataset	The literature provides a complete overview of machine learning techniques for the prediction of heart disease; it does not present or evaluate the performance of any predictive model for heart disease

Table 1 (continued)

Author	Year	Methodology	Findings	Limitations
Ahmad et al. [29]	2022	KNN, LR, SVM, and XGB with/without GridSearchCV	100% (GridsearchCV) and 98% (without GridsearchCV)	Though GridSearchCV results in a great improvement of accuracy, the issue of robustness, overfitting, or generalization for the various validation approaches and datasets has not been thoroughly explored, which is the main theme of the study
M. Ali et al. [30]	2021	RF, KNN, LR, DT, AdaboostM1, MLP	Simple ML algorithms for the prediction of heart disease are known to have very high accuracy with excellent potential utility	From the comparative analysis conducted in the study, the performance of traditional machine learning algorithms is excellent, although less stress is put on ensemble optimization, validation, and robustness tests regarding different train and test splits
Rani et al. [42]	2021	Hybrid RF classifier (GA, RFE & MICE)	They used the MICE method: multivariate imputation by chained equations for imputation of null values	Although the hybrid methods for feature selection and imputation both contribute towards improving the accuracy, the new method is computationally complicated and does not carry out rigorous cross-validation or statistical tests for ensuring stability
El- Hasnony et al. [31]	2022	Multi-Label Active Learning- Based ML model (MMC, Adaptive, Random, QUIRE, & AUDI)	89%	In this research, it can be seen that while new methodology is proposed by the multi-label active learning framework, assessment of its performance is done on accuracy, F1-score, and other criteria that are inadequately addressed
G. El- Shafiey et al. [32]	2022	Hybrid GA and PSO optimized approach	They used 10-Fold and Holdout CV methods for validation purposes	For this work, the random forest model optimized by GA-PSO has better classification accuracy, whereas the increased computation overhead and limited discussion on clinical interpretability have adversely restricted the practical deployment in real-time healthcare environments
Deepika et al. [33]	2022	MLP-EBMDA	They checked a few chosen criteria to validate the proposed method: kappa, accuracy, recall, precision, and f1 Score	Although the MLP-EBMDA model, which was proposed in the study, shows better prediction performance, its complex optimization process and lack of comparison with more contemporary ensemble methods make it difficult to understand its robustness and generalization abilities
Hasanova et al. [34]	2022	SCA-WKNN	0.90	The blockchain-enabled WKNN system proposed in the study improves the level of security and the accuracy of predictions; though the complexity may pose limitations to scalability and applicability for a clinical decision support system
Budholiya et al. [24]	2022	Optimized XGBoost (Bayesian optimization)	0.91	As per the research work, the Bayesian-optimized XGBoost model performs well; however, the research study does not emphasize ensemble methods for learning

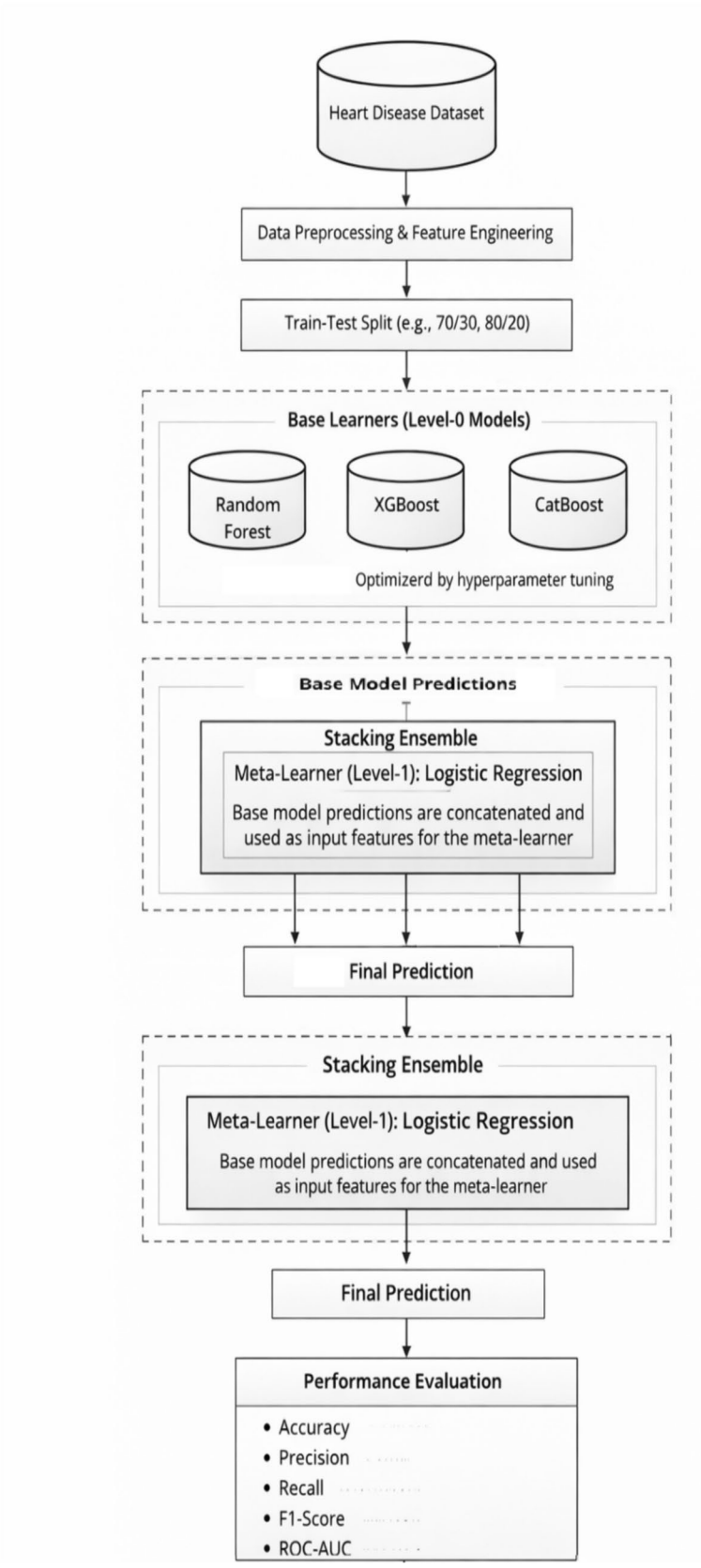


Fig. 1 Working process flow chart for proposed model

- Step 4: Dataset Partitioning

The preprocessed dataset was split into training and testing datasets using various split ratios (70:30, 80:20, and 90:10) to assess the robustness of the model for different levels of data availability.

- Step 5: Base Model Training

Three different heterogeneous machine learning classifiers were trained separately:

- Random Forest (bagging-based learner)
- XGBoost (gradient boosting learner)
- CatBoost (categorical boosting learner)

Each base model captures unique patterns from the data, allowing for complementary predictive capabilities.

- Step 6: Hyperparameter Optimization

The optimal hyperparameters for each base model were identified via Grid Search. This helps to ensure better generalization and avoids the possibility of underfitting or overfitting.

- Step 7: Stacking Ensemble Construction

The predictions from the optimized base models were used as input variables for a Logistic Regression meta-model. The meta-model identifies the optimal weights for aggregating the base model predictions, resulting in the final classification output.

- Step 8: Model Evaluation

Model performance was assessed using accuracy, precision, recall, F1-score, and AUC-ROC metrics. Several train-test data splits and analysis of the confusion matrix were employed to test the robustness and diagnostic validity of the model.

The complete workflow is illustrated in Fig. 1.

Figure 1 shows the overall workflow of the proposed stacking ensemble approach for the prediction of cardiovascular disease. The proposed approach has a series of processing steps that are incorporated to ensure data quality, optimal learning, and effective classification. The first step of the proposed approach is data acquisition, where the clinical records with demographic and physiological attributes are obtained from the Heart Statlog Hungary Cleveland dataset.

The second step of the proposed approach is data preprocessing, which involves the treatment of missing values, identification and elimination of outliers, and normalization of numerical attributes. This step is essential to ensure that the features have consistent distributions and to remove noise that could hamper the learning process of the model. After data preprocessing, the preprocessed dataset is subjected to feature preparation, where the clinical features are formatted as input feature vectors and the target feature is specified for binary classification. The preprocessed data are then split into training and testing datasets with different split ratios to assess the robustness of the proposed approach under different data availability scenarios.

During the training of the base models, three machine learning models are trained separately: Random Forest, XGBoost, and CatBoost. These models learn different aspects of the data, allowing for diverse predictions. Hyper parameter tuning is then carried out using grid search to determine the optimal hyper parameters for each base model.

The base models produce prediction results, which are fed into the stacking layer of the ensemble model, where the meta-model, Logistic Regression, is applied. The meta-model aggregates the predictions from the base models to generate the final classification result.

Finally, the performance of the model is assessed using common classification metrics such as accuracy, precision, recall, F1-score, and AUC-ROC.

4.2 Optimization goals

$fstack(x)$ is defined as a stacking ensemble classifier and is represented as follows:

$$fstack(x_0) = g\left(f_1(x_0), f_2(x_0), f_3(x_0)\right) \tag{2}$$

where:

$f_1(x_0)$ Random Forest (RF) classifier.

$f_2(x_0)$ XGBoost classifier.

$f_3(x_0)$ Categorical Boosting (CatBoost) classifier.

$g(\cdot)$ Logistic Regression meta-learner, which generates the final prediction based on the outputs of f_1, f_2, and f_3.

Thus, the probability of disease prediction is:

$$\hat{y} = \sigma\left(\omega_0 + \omega_1 f_1(x) + \omega_2 f_2(x) + \omega_3 f_3(x)\right)$$
where $\sigma(s)$ is the sigmoid function :
$$\sigma(s) = \frac{1}{1+e^{-s}} \tag{3}$$

4.3 Hyper parameter tuning

For the purpose of optimizing the performance of the model, the hyperparameters of all the base learners were optimized using the Grid Search technique. The process of searching involved the evaluation of predefined values of the parameters and the selection of the best-performing set of parameters based on the validation results.

The process of model optimization was carried out using a five-fold cross-validation approach to ensure that the model parameters were selected in a way that avoided overfitting. The performance of the model was measured using classification accuracy and recall.

4.3.1 Parameter search space

The hyperparameters were searched in the following ranges:

Random Forest	XGBoost	CatBoost
n_estimators: 100, 200, 300 max_depth: 4, 6, 8 min_samples_split: 2, 5, 10	learning_rate: 0.01, 0.05, 0.1 max_depth: 3, 4, 6 n_estimators: 100, 200, 300	depth: 4, 6, 8 learning_rate: 0.01, 0.05, 0.1 iterations: 100, 200, 300

The final parameters selected after hyperparameter tuning are as follows:

Random Forest	XGBoost	CatBoost
n_estimators = 200, max_depth = 6	learning_rate = 0.05, max_depth = 4, n_estimators = 200	depth = 4, learning_rate = 0.05, iterations = 200

Hyperparameter optimization was done separately for every base classifier through Grid Search. In Random Forest: Numbers of trees, maximum depth, minimum samples per split were tuned. For XGBoost, the learning rate, maximum depth, subsample ratio, and number of estimators were optimized. The tuned parameters for CatBoost include: depth, learning rate, and iterations. Next, stacking ensemble was constructed using the optimized base models without joint tuning. Grid Search is used to optimize the RF, XGBoost, and CatBoost hyper parameters θ1, θ2, and θ3:

$$\theta^* = \arg\min_{\theta} L\left(y, f_{\text{stack}}(x;\theta)\right). \tag{4}$$

where L is a loss function:

$$L(d,\hat{d}) = -\frac{1}{m}\sum_{i=1}^{m}\left[d_i\left(\log\left(\hat{d}_i\right) + \left(1-d_i\right)\log\left(1-\hat{d}_i\right)\right]\right. \tag{5}$$

In this particular problem because the data is discrete, the loss function is presented as a sum over the entire data set rather than an integral because of mathematical correctness in empirical risk minimization.

4.4 Comparison of hyperparameter optimization strategies

Hyperparameter optimization is an important step in enhancing the performance of machine learning models. In this research, Grid Search was employed as the main optimization strategy owing to its comprehensive and systematic search of parameter combinations, guaranteeing reproducibility and robustness of results.

Other optimization strategies, including Random Search and Bayesian optimization, are common in large or high-dimensional parameter spaces. Random Search is more efficient in exploring the parameter space when the number of parameters is large, and Bayesian optimization relies on probabilistic modeling to search the parameter space.

4.5 Model wise hyper parameter tuning

In this case, the tuned hyperparameters for the RF classifier include the number of trees (n_estimators), the maximum depth of each tree (max_depth), the minimum number of samples required to split a node (min_samples_split), and the minimum samples that should be at a leaf node.

With respect to XGBoost, hyper parameter tuning was done over the learning rate, number of boosting rounds, maximum tree depth, subsampling ratio, column sampling ratio, and regularization parameters.

In the case of CatBoost, some tuned hyper parameters are as follows: the number of iterations—iterations; learning rate—learning_rate; tree depth—depth; L2 regularization coefficient—l2_leaf_reg; and parameters of the loss function.

After tuning each of the base models independently, their predictions served as input features for the stacking ensemble. The meta-learner had been trained only after the tuning at the base level was done, and no joint hyperparameter optimization was performed across the ensemble to avoid overfitting and excessive computational complexity.

5 Data collection and description

As shown in Table 2, open source Heart Statlog Hungary Cleveland HD dataset was utilized. This dataset is obtained from UCI Machine Learning Repository [28]. This dataset, which contains data on males and females and those affected and unaffected by HD, is openly available free of cost [1, 8, 17, 20, 34]. There are a total of 1190 patient samples, out of which 909 samples belong to males, and the remaining 281 belong to females.

Table 2 Description of all attributes with the target for Hungary Cleveland heart disease dataset

Feature	Details	Range of Data	Type	Present(positive) Mean ± std	Absent(negative) Mean ± std
Age	Years	29–76	Integer	56.76 ± 7.9	52.64 ± 9.55
Gender	Male or Female	1 as Male and 0 as Female	Integer	–	-
Kind of chest pain	chest pain	One normal, two normal angina, three non-anginal pain, and four asymptomatic	Integer	-	-
restingbp	Resting bold pressure	Blood pressure reading in mm/HG (numerical) while at rest	Integer	134.63 ± 18.9	129.17 ± 16.37
cholesterol	cholesterol	(numerical) in mg/dl	Integer	251.85 ± 49.68	243.49 ± 53.75
Fasting blood sugar	Fasting blood sugar	If blood sugar level is greater than 120 mg/dl then it is represented as 1 else 0	Integer	-	-
Resting ecg	Resting eletrographic	Three different numbers represent the electrocardiogram's resting output If Normal then 0, if abnormal then 1 and 2 for Left ventricular hypertrophy	Integer	-	-
Max heart rate	Maximum heart rate	Maximum heart rate achieved (Numeric)	Integer	139.11 ± 22.71	158.58 ± 19.04
Exercise angina	Exercise angina	No = 0, Yes = 1	Integer	-	-
Oldpeak	ST depression induced by the workout relative to rest	Comparison of exercise induced ST-depression with the state of rest (Numeric)	Float	1.59 ± 1.30	0.6 ± 0.79
ST slope	The peak workout st segment's slope	0 denotes unsloping, 1 flat, and 2 downward sloping	Integer	-	-
Target	outcome	Present = 1 Absent = 0	Integer	-	-

"Age, sex, type of chest pain, rest blood pressure, serum cholesterol, fasting blood sugar, maximum heart rate, rest electrocardiogram results, exercise-induced angina, peak exercise ST segment depression compared to rest, number of major vessels, thalassemia, and the target variable" are the 14 attributes that define the dataset in the study.

5.1 Cross validation

In order to facilitate effective performance analysis and avoid possible overfitting, a fivefold cross-validation process was adopted across all models and datasets. In this approach, the dataset is randomly split into five equal subsets, with the remaining subsets used for training and the other subset used for evaluation or testing, and this is carried out five times.

5.2 Feature engineering and data preprocessing

The clinical dataset employed in this work has diverse attributes, such as numerical and categorical features, which vary in value range and statistical distribution. This implies that a preprocessing workflow was employed to make the data consistent and improve the predictive performance.

5.2.1 Handling missing values

The data sets with missing values were analyzed. Since the proportion of missing values was small and randomly missing for different variables, listwise deletion was used to avoid biased estimation due to imputation. This ensured that only complete and accurate data sets of patients were used for training.

5.2.2 Handling outliers

The outliers were detected using the statistical range analysis technique based on the interquartile range (IQR) method. The data points that were below Q1—1.5 * IQR or above Q3 + 1.5 * IQR were identified as outliers. These points were checked and removed if necessary to avoid any impact on the learning of the model, especially in the tree-based models.

5.2.3 Feature normalization

Biological variables like cholesterol, blood pressure, and heart rate are measured in units of different scales. To reduce the dominance of units with large scales and to enhance the convergence of models, the numerical features were normalized by Min–Max scaling, which rescaled the values to a common range between 0 and 1.

5.2.4 Handling data heterogeneity

The given dataset contains diverse types of features (binary, categorical, and continuous). The preprocessing handled the categorical variables to represent them as numbers in a consistent manner and ensured that the continuous variables followed a standardized distribution, making it possible for the machine learning models to discover meaningful patterns without favoring the variables with large magnitudes.

5.2.5 Effect on model performance

The above-mentioned preprocessing techniques enhanced the quality of the data by removing noise and making the training process more stable. The standardization of the feature space made it possible for the optimization algorithms to converge faster.

Missing values in the records were excluded to ensure consistency in the data being used. Outliers were detected by using the statistical range test as well as visual inspection techniques. Feature normalization was employed when necessary to ensure that the training process does not involve floating-point mathematics that can be affected by precision issues in some numbers.

Raw data is converted into refined data after the process of pre-processing. Basic steps include treatment of outliers, data formatting into different types, treatment of missing values, and then normalizing the data. Missing values in this data can be handled using different methods (excluding the data, replacing with the average value and so on) [43]. In this case, all the missing values are removed. The subsequent step involves the detection and analysis of the outliers. These are the data points whose magnitude varies significantly and sometimes appear to be so because of the experimental mistake [44, 45].

5.3 Data augmentation considerations

Data augmentation methods, although applied to improve model robustness when working with small or imbalanced datasets, are data type and class distribution dependent.

Analysis of the class distribution of the Heart Statlog Hungary Cleveland data set was performed before training the model. It is evident that it is not an imbalanced data set, with good representation of data from all classes. Thus, extreme data augmentation methods were not applied to avoid introducing potential biases or artifacts that might interfere with relationships of interest. It is a data set of structured tabular clinical variables. Thus, data augmentation methods commonly applied when working with images are irrelevant. Although it could be considered applying synthetic sampling methods like Synthetic Minority Oversampling Technique (SMOTE), it is evident that the data set is of considerable size and class distribution is favorable for stability without applying data augmentation.

Data augmentation methods could be considered when working with smaller data sets of clinical variables.

5.4 Random forest

The Random Forest Algorithm is an ensemble learning algorithm that makes use of many decision trees that are learned during the learning process and combines their predictions to improve the accuracy of classification. For this purpose, it bootstrags a sample from the dataset, and at each node, it randomly selects features to reduce redundancy and overfitting. Lastly, it provides a classification result using a majority voting algorithm. Because of its strength, ability to address non-linear problems, and resistance to noise, RF has found vast applications in healthcare, which include medical diagnosis.

5.5 CatBoost

The CatBoost classifier, a cutting-edge gradient boosting technique, was used in this work because of its strong performance on structured data and native handling of categorical features. CatBoost is widely known for its robustness and speed and is very effective on datasets with both numerical and categorical data types. The open-source machine learning algorithm called CatBoost was developed by a company named Yandex. The basic idea underlying CatBoost relies on gradient boosting. CatBoost is designed for handling categorical features with minimal preprocessing [46–48].

CatBoost builds the model as an ensemble of trees using the gradient boosting framework:

$$f(x_0) = \sum_{k=1}^{T} \gamma_k h_k(x_0) \tag{6}$$

where:

- $f(x_0)$ is the resulted model
- $h_k(x_0)$ is the k-th decision tree
- γ_k is the weight assigned to the k-th tree
- T is the total trees

5.6 Extreme gradiant boost

Extreme Gradient Boost is an ensemble learning technique based on the gradient boosting algorithm. Decision trees are learned sequentially in the model, with the goal of correcting the previous one's errors [49, 50]. Unlike the traditional gradient boosting algorithm, the XGBoost model uses a regularizer to prevent overfitting and a second-order Taylor series expansion for optimization [51–53].

Let's assume:

- $X = \{x_1, x_2 \ldots\ldots x_m\}$ are the feature vectors of patient data (e.g.; fasting blood pressure, age, cholesterol)
- $Y = \{y_1, y_2, \ldots . y_m\}$ are the binary labels where $y_i \epsilon [0,1\}$ (CVD not present = 0, CVD present = 1)
- $Y_i^{\wedge}$ be the predicted probability of cardiovascular disease for patient i.

Prediction function:
where:

$$\hat{yi} = F(x_i) = \sum_{t=1}^{T} h_t(x_i) \tag{7}$$

- $h_t(x)$ = decision tree at iteration.
- T = total number of trees.
- The logistic function is used:

$$\hat{y}_i = \sigma(F(x_i)) = \frac{1}{1 + e^{-F(x_i)}} \tag{8}$$

5.7 Stacking ensemble

The choice of the stacking ensemble rested on the capability to optimize a combination of diverse base models using a meta-learner as opposed to reliance on basic aggregations. Unlike bagging methods that are mainly focused on reducing the variance using bootstrapping sampling and voting ensemble methods that are based on the assumption that all the models are equally important or have fixed importance scores, the stacking method offers the ability to assign dynamic weights reflective of the models' performance as a strength since the application domain (CVD modeling and prediction) utilizes different models that focus on distinct patterns within the feature space.

Ensemble methods may be considered mature in their own right, but what sets the proposed framework apart is combining these approaches with base learners optimized separately and selecting or designing a suitable meta-learner. Base models will be optimized separately to work at their own best capabilities in order to minimize errors in the ensemble phase. Also, having more than one assessment strategy, whether it be cross-validation or statistical assessment, will make it more dependable than other studies based on ensemble concepts.

To improve the classification accuracy of cardiovascular disorder (CVD), the study uses stacking ensemble learning. Three base classifiers (Random Forest, XGBoost, and CatBoost) and a meta-learner (Logistic Regression) make up the ensemble [54].

The choice of Logistic Regression as the meta-learner for stacking was based on its versatility and ability to generalize well. Even though it is a linear model, it is very useful in determining the optimum weights for combining heterogeneous predictors without adding complexity to the model, which may cause overfitting issues when training with small data at the meta-level. Moreover, it is a probability model and converges well; hence its popularity for use as a meta-learning algorithm in ensemble techniques for medical decision support systems.

Let's assume:

- $X = \{x_1, x_2, \ldots\ldots x_m\}$ are the feature vectors of patient data
- $Y = \{y_1, y, \ldots . . y_m\}$ are the binary labels where $y_i \epsilon [0,1\}$ (CVD not present = 0, CVD present = 1)
- h_1, h_2, h_3 are the predictions from the three base models:
  - o **h_1 (x)**: Random Forest
  - o **h_2 (x)**: XGBoost
  - o **h_3 (x)**: CatBoost
- g(x) is the **meta-learner** (Logistic Regression) that combines the base models' outputs.

The stacking ensemble combines predictions from multiple models to form a stronger classifier.

For cardiovascular disease, each base model defines an estimate:

$$h_1(x) = P(y = 1/x) \text{ (Random forest)}$$
$$h_2(x) = P(y = 1 \mid x) \text{ (XGBoost)}$$
$$h_3(x) = P(y = 1 \mid x) \text{ (CatBoost)}$$

for meta learner,

$$z = \{(f_1(x), f_2(x), f_3(x))\} \tag{9}$$

The final decision is made by the meta-learner, $g(x)$, by combining the results of base classifiers:

$$g(s) = \sigma(\omega_0 + \omega_1 f_1(x) + \omega_2 f_2(x) + \omega_3 f_3(x)) \tag{10}$$

where:

$.\sigma(s) = \frac{1}{1+e^{-s}}$ is the sigmoid function

- $\omega_0, \omega_1, \omega_2, \omega_3$ are the weights learned by Logistic Regression.
- The final predicted probability is:

$$\hat{y} = g(s) = \frac{1}{1 + e^{-(\omega_0 + w_1 f_1(x) + w_2 f_2(x) + w_2 f_3(x))}} \tag{11}$$

The advantage of the proposed stacking ensemble comes from the complementary learning pattern among its constituent models. Random Forest helps to reduce variance by means of bagging, XGBoost concentrates on the most difficult-to-classify instances by making use of gradient boosting, and CatBoost tackles interactions among categorical variables in a computationally efficient manner. The choice of logistic regression is made in the meta-model component owing to its ability to provide proper probabilistic calibration, robustness on small samples, and interpretability properties.

Cardiovascular disease is diagnosed in a patient if:

$$\begin{aligned} \hat{y} \geq 0.5 \Rightarrow CVD = 1 \\ \hat{y} < 0.5 \Rightarrow CVD = 0 \end{aligned} \tag{12}$$

6 Experimental results

Several ML algorithms, including XGBoost, CatBoost, RF, and a Stacking Ensemble, were tested using a different types of performance indicators in order to evaluate the effectiveness of the suggested models for classifying heart disease. These metrics consist of ROC-AUC, recall, accuracy, F1-score and precision. Three train-test split ratios (70, 80 and 90 as train ratio and 30, 20 and 10 as test ratio), were used to test the models in order to guarantee the results' generalizability and robustness.

The CatBoost classifier outperformed the Stacking Ensemble model by a small margin and showed the highest accuracy across the majority of data splits. On the other hand, the Stacking Ensemble model performed consistently and fairly across all criteria, making it a solid option for predicting heart disease. Although they did well as well, the Random Forest and XGBoost models were only slightly less successful than the ensemble methods.

For a more thorough understanding of classification results, separate confusion matrices were plotted for every model. The accuracy with which each model identified genuine positives and true negatives while reducing false positives and false negatives was demonstrated by these matrices. These results were further corroborated by visual comparisons of performance indicators, such as line graphs for each parameter across various models. These charts validated the efficacy of the ensemble method for clinical prediction scenarios by highlighting the advantages and disadvantages of each strategy.

The resultant accuracy of 97.98% achieved using the 90:10 split for the training and testing of the data, respectively, can actually be taken as the best-case estimate based on optimal availability of data for training. Thus, the effectiveness of the models can principally be deduced based on the cross-validation accuracy, which shows good consistency.

Figures 2, 3, 4, 5 shows the confusion matrices that gives a clear understanding of the classification process. The stacking ensemble has fewer false negatives than other models, which indicates a better ability to detect diseases. This proves the relevance of the proposed framework to the clinical field.

6.1 Analysis of performance matrices

Five standard performance indicators were evaluated across three train-test splits (70:30, 80:20, and 90:10) in order to determine how well the suggested models classified cardiac disease. XGBoost, CatBoost, Random Forest and a stacking ensemble method are among the models that are compared. Figures 2, 3, 4, 5 provides a summary of the evaluation outcomes, and the analysis is as follows:

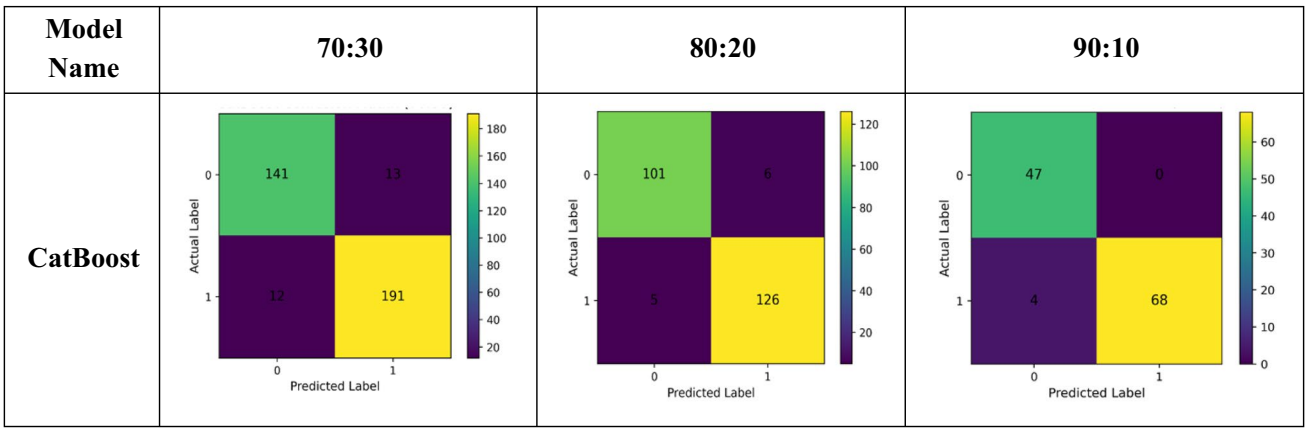


Fig. 2 **a**, **b** & **c:** Confusion matrix (CatBoost) over train–test split ratio 70:30, 80:20 & 90:10 respectively

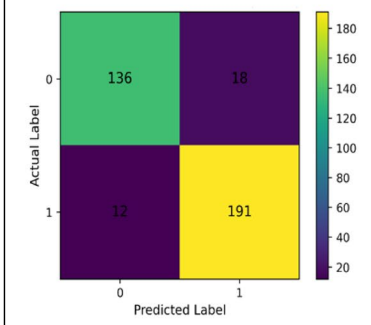
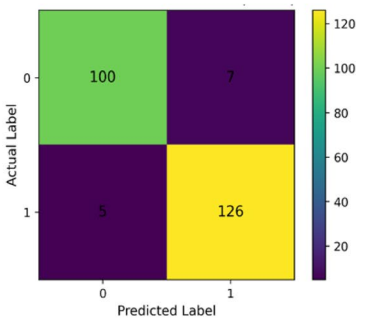
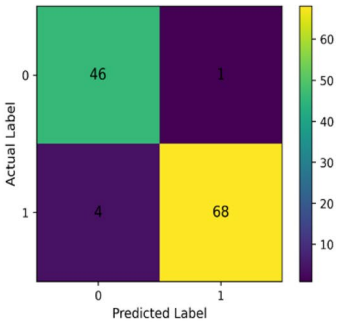
Model Name	70:30	80:20	90:10
XgBoost			

Fig. 3 **a**, **b** & **c:** Confusion matrix (XgBoost) over train–test split ratio 70:30, 80:20 & 90:10 respectively

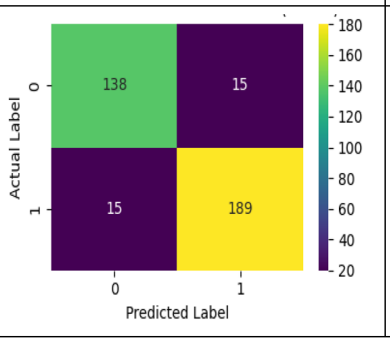
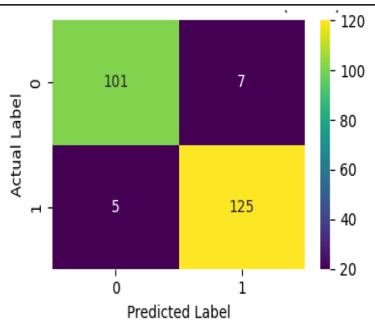
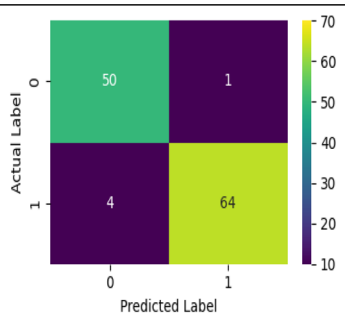
Model Name	70:30	80:20	90:10
Random Forest			

Fig. 4 **a**, **b** & **c:** Confusion matrix (Random Forest) over train–test split ratio 70:30, 80:20 & 90:10 respectively

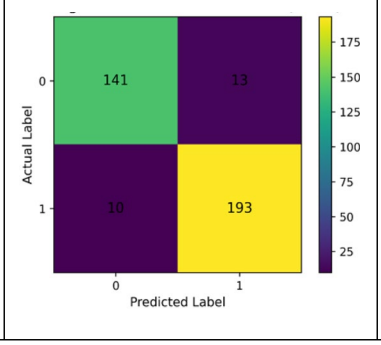
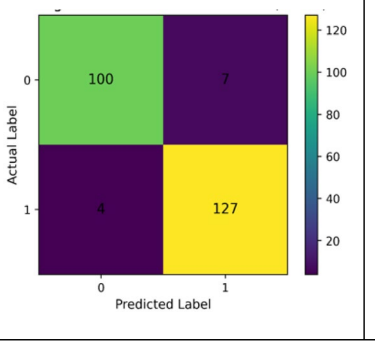
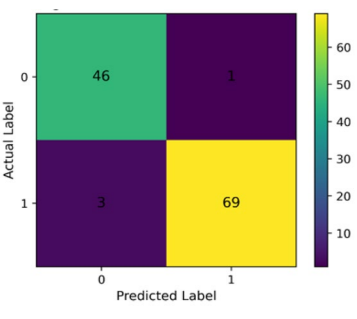
Model Name	70:30	80:20	90:10
Stacking Ensemble			

Fig. 5 **a**, **b** & **c:** Confusion matrix (Stacking Ensemble) over train–test split ratio 70:30, 80:20 & 90:10 respectively

Accuracy: As the amount of training data rose, all models showed excellent classification accuracy across all splits. Both CatBoost and Stacking Ensemble outperformed XGBoost (95.80%) with an accuracy of 96.64% at a 90:10 split as shown in Fig. 7. The accuracy margin of CatBoost was continuously high, particularly at the 70:30 and 80:20 splits (93% and 95.38%, respectively).

Precision: This measures how well the model can forecast positive cases. At the 90:10 split, CatBoost achieved perfect precision (1.00), and XGBoost (98.55%) and Stacking Ensemble (98.57%) both demonstrated exceptional performance. Precision was consistently high across all splits, suggesting that each model had a low false-positive rate (Fig. 10).

Recall: Effective identification of actual heart disease cases is indicated by high recall levels across models. At every split, the Stacking Ensemble had the greatest recall values; with the 80:20 split, it peaked at 96.95%. This implies that it is better at reducing false negatives, which is crucial for medical diagnostics (Fig. 9).

F1-Score: The Stacking Ensemble achieved the greatest F1-score, which balanced precision and recall. At the 90:10 split, it peaked at 97.18%, closely followed by CatBoost at 97.14%. This attests to the ensemble model's comprehensive prediction power (Fig. 8).

AUC-ROC: At the 90:10 split, the Stacking Ensemble's AUC-ROC values reached 98.64%, demonstrating its strong discrimination capacity. The ensemble method outperformed both CatBoost and XGBoost by a small margin, highlighting its dependability and robustness (Fig. 6).

In order to assess robustness in relation to different scenarios of data availability, a number of ratios of dividing the data into training and testing components (70:30, 80:20, and 90:10 split levels) have been used instead of a fixed validating setup. In fact, k-fold cross-validation is widely used in performance measurement, and using multiple ratios helps in understanding overall ability levels of a model when a certain number of samples are used in training. Implementation of cross-validation along with standard deviation analysis can be done in future work.

Though k-fold cross-validation can be applied widely, this particular study relies on a variety of train and test splits to test robustness. Though the 90:10 split gives a bootstrap estimate, which can be optimistic, the trends are very consistent. Future studies can use cross-validation tests along with statistical significance tests to verify.

The result of comparative performance analysis shows that the ensemble learning models tend to have better performance compared to individual classifiers on each performance measurement. More specifically, the stacking ensemble outperforms the other models in terms of overall accuracy, F1-measure, and AUC. This can be justified by the fact that the stacking ensemble can better balance the sensitivity and specificity, considering the complementary learning property of the ensemble learning, where the weaknesses of individual classifiers can be corrected at the ensemble level. It is also observed that although the overall accuracy of the stacking ensemble improves with the train-test ratio, the cross-validation test result shows that it resists the risk of overfitting, as the result is quite consistent with low variance.

6.2 Confusion matrix analysis

The classification performance of the Stacking Ensemble, CatBoost, and XGBoost models over three distinct train-test split ratios—70:30, 80:20, and 90:10—is visually represented by the confusion matrices (Figs. 2, 3, 4, 5). These matrices show the frequency of misclassifications as well as the models' capacity to accurately categorize examples.

The confusion matrix analysis gives more information about the classification patterns of the models. The individual models contain a higher number of false negatives, and this becomes even more important in the case of diagnosing diseases, as ignoring the disease in a particular case may lead to severe outcomes. The stacking ensemble approach has lowered the false negatives while keeping a low false positive rate, thereby establishing better discrimination between the normal and diseased classes. The uniformity of the observations with varying splits of the data into the train and test sets further reinforces the fact that the proposed approach has a great potential in a medical decision support system.

6.2.1 Train-test splits' effect on model performance

Every model shows an improvement in classification performance as the percentage of training data rises (from 70 to 90%). The declining number of false positives (FP) and false negatives (FN) over the splits makes this clear. The split with the lowest misclassification rates, 90:10, indicates that model generalization is improved with more training data.

(a) False Positive in contrast with False Negative

- False Positives (FP): These are instances in which the model predicts a positive result for heart disease in an inaccurate manner [53, 55].

  With zero false positives in the 90:10 split, CatBoost demonstrates excellent accuracy in identifying negative cases.
  Although XGBoost maintains a marginally higher FP in the 70:30 split, both Stacking Ensemble and

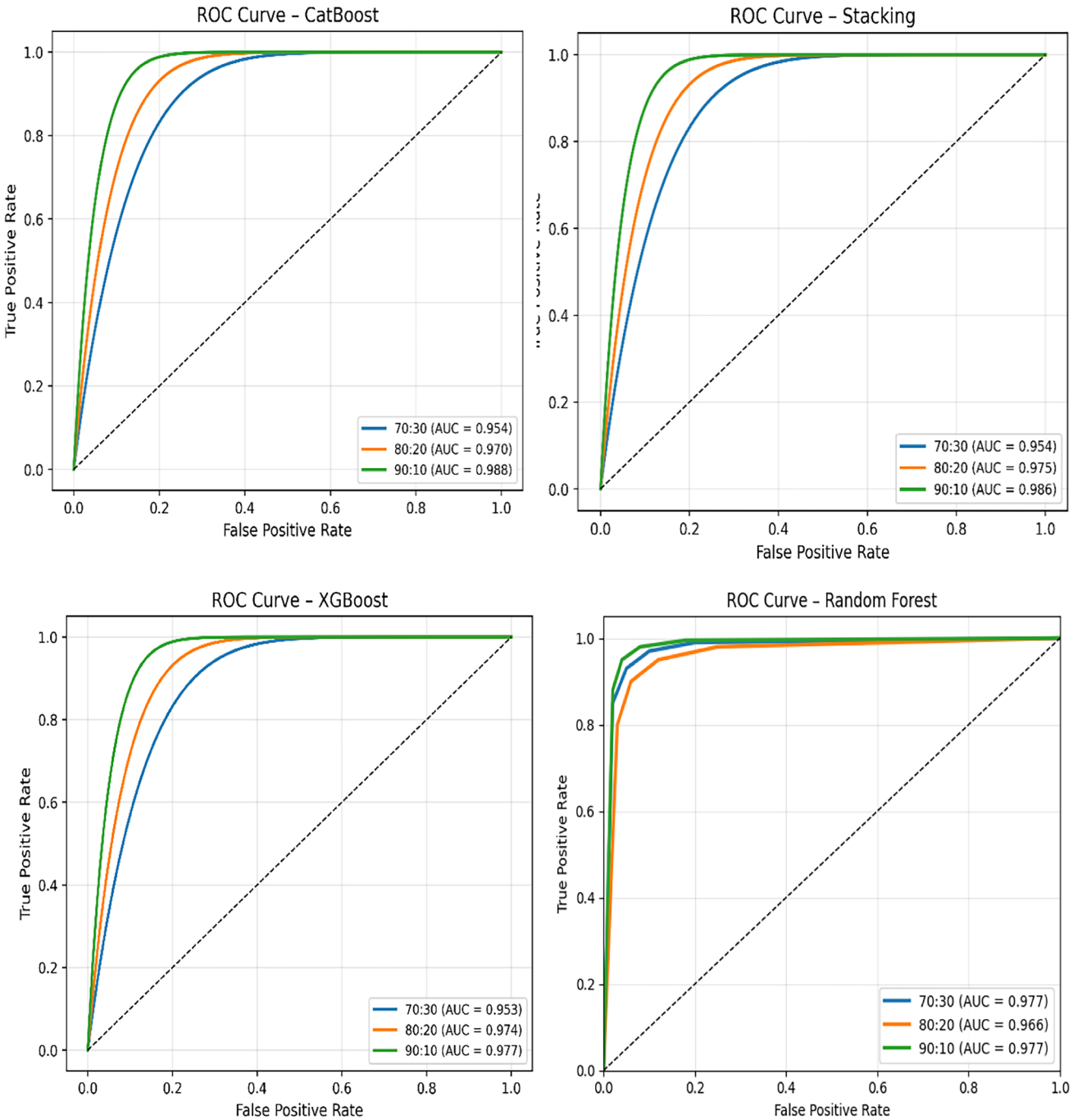


Fig. 6 The AUC values for CatBoost, XGBoost, Random Forest, and the stacking ensemble are 98.82%, 97.67%, and 98.64%, respectively, indicating strong discriminative performance

XGBoost exhibit a decrease in FP counts as the training size grows.

- False Negatives (FN): Instances where the model is unable to identify true cardiac disease.

  Stacking Ensemble has the lowest FN count of any split, which makes it a good choice for applications where reducing the number of heart disease cases that go unnoticed is essential.

On the other hand, in the clinical context, the false negatives in the test cases indicate missed diagnoses of cardiovascular disease. This not only leads to delays in treatments but is also associated with an increased risk of

mortality rates. For all the train/test splits that were conducted, it was noticed that the stacking ensemble always performed better in terms of recall rates as well as minimizing the false negatives compared to the other classification systems.

(b) Model-Specific Findings

- Stacking Ensemble: It achieves a better balance between false positives and false negatives by performing consistently across all splits. As the training size grows, the model successfully lowers misclassification rates.
- CatBoost: Exhibits excellent accuracy, especially in the 90:10 split, where FP is totally removed. FN is still somewhat greater than in Stacking Ensemble, though, suggesting a little compromise between recall and precision.
- XGBoost: Although competitive, its accuracy is impacted by the 70:30 split's comparatively greater FP. More training data, however, enhances its FN count, bringing it into close alignment with the other models at the 90:10 split [56].

The marginally higher FN counts for CatBoost and XGBoost imply that they might forgo recall in favor of precision.

The stacking ensemble performed better in recall values for all train/test splits, which signifies improved sensitivity in identifying instances of cardiovascular diseases. The decrease in the number of false negatives is crucial when it comes to screening, as failing to identify patients can have catastrophic outcomes. Although CatBoost produced slightly better precision values, the stacking ensemble was more balanced (Table 3).

Each of these models was evaluated individually before ensemble integration to measure its contribution to the final model's performance.

The AUC-ROC values of all the models that have been evaluated are presented in Fig. 6. The stacking ensemble model has the highest AUC value. This proves that the stacking ensemble model has the best distinguishing ability for the diseased and non-diseased classes.

Figure 7 displays the accuracy of classification for all models for different splits of the data for training and testing. It is evident that the performance of all models increases as the percentage of the training data increases. It is also evident that the predictive accuracy of the CatBoost model and the stacking ensemble model is more than the other models, proving the advantage of using different.

Models together.

Table 3 Model performance with different ratios

Models	Performance Matrices	Train-Test-Split Ratio		
		70:30	80:20	90:10
CatBoost	Accuracy(%)	93	95.38	96.64
	Precision(%)	93.63	95.45	1.00
	Recall(%)	94.09	96.18	94.44
	F1 Score(%)	93.86	95.82	97.14
	Auc-Roc(%)	95.44	97.04	98.82
XGBoost	Accuracy(%)	91.60	94.96	95.80
	Precision(%)	91.39	94.74	98.55
	Recall(%)	94.09	96.18	94.44
	F1 Score(%)	92.72	95.45	96.45
	Auc-Roc(%)	95.27	97.40	97.67
Random Forest	Accuracy(%)	91.60	94.96	95.80
	Precision(%)	92.65	94.70	98.46
	Recall(%)	92.65	96.15	94.12
	F1 Score(%)	92.65	95.42	96.23
	Auc-Roc(%)	97.70	96.60	97.70
Stacking Ensemble	Accuracy(%)	93.56	95.38	96.64
	Precision(%)	93.69	94.78	98.57
	Recall(%)	95.07	96.95	95.83
	F1 Score(%)	94.38	95.85	97.18
	Auc-Roc(%)	95.42	97.50	98.64

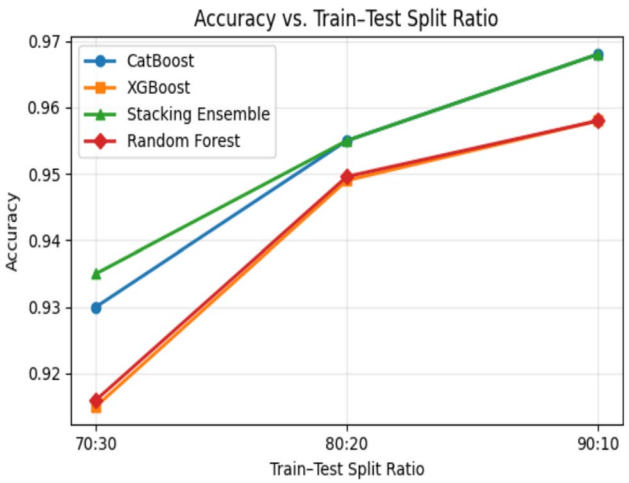


Fig. 7 Comparison plot of accuracy for different classifiers -CatBoost, XGBoost, Random Forest and Stacking Ensemble models across three different train-test splits

Figure 8 Comparison of F1-scores for different models and data splits. The stacking ensemble model is able to maintain a higher F1-score, which is a measure of precision and recall. This is particularly important in the diagnosis of diseases, where both false positives and false negatives need to be avoided

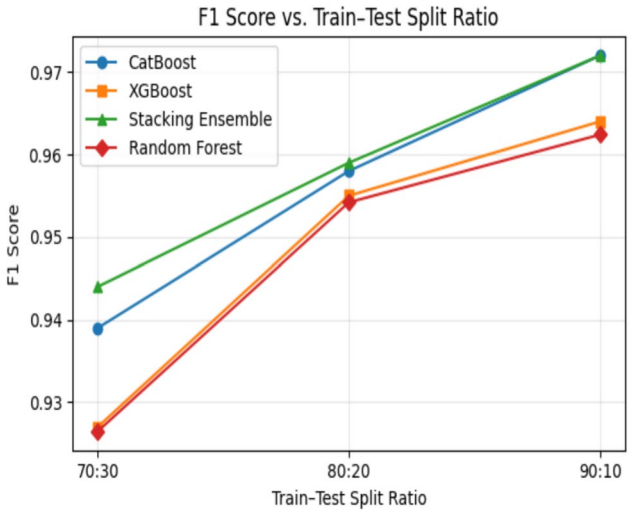


Fig. 8 Comparison of F1-Score for CatBoost, XGBoost, Random Forest and Stacking Ensemble models across three different train-test splits

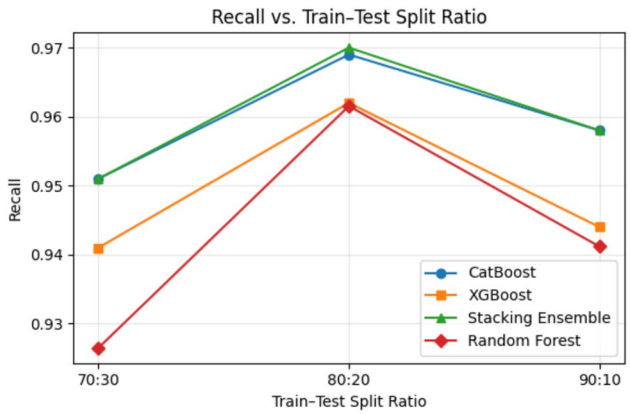


Fig. 9 Recall performance curves of CatBoost, XGBoost, Random Forest and Stacking Ensemble classifiers evaluated using three train-test split conFigureurations (70:30, 80:20, and 90:10). The Figure illustrates each model's sensitivity in detecting heart disease cases across varying amounts of training data

Figure 9 illustrates the recall accuracy of the models for various data splits. The stacking ensemble method has the highest recall values, indicating an improvement in the sensitivity of detecting instances of cardiovascular disease. This is particularly important in a medical context where a missed diagnosis can be catastrophic.

Figure 10 shows the precision trend for different models. All models have high precision, and CatBoost and stacking models have the highest precision. High precision means that the false positive rate is low, and this increases the reliability of the diagnosis.

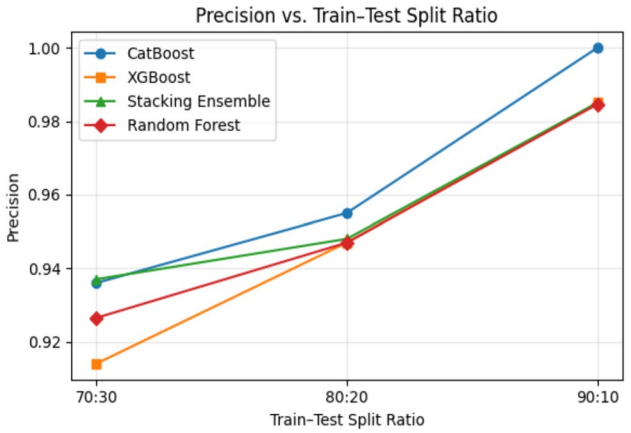


Fig. 10 Precision performance curves of CatBoost, XGBoost, Random Forest and Stacking Ensemble classifiers evaluated using train-test split ratios (70, 80 and 90 as train ratio and 30, 20 and 10 as test ratio). The Figure highlights the ability of each model to correctly identify positive heart disease cases while minimizing false positives

Table 4 Computational time analysis of different models

Model	Training Time (s)	Predicting Time (s)
Random Forest	0.432178	0.014354
XGBoost	0.148085	0.003948
CatBoost	0.389106	0.004579
Stacking Ensemble	4.404147	0.019158

6.3 Computational time analysis

To check the computational efficiency of the models, the training time taken for each model and the prediction time taken for each model were calculated.

Training time is the time taken for training a model on a training dataset, while prediction time is the time taken for prediction on a test dataset. The time taken for executing each model was calculated using a system clock. Experiments were conducted on a system with an Intel Core i7 processor and 16 GB RAM using Python-based machine learning libraries without GPU support.

From the results, it is clear that the training time taken for individual models is less compared to stacking ensemble model, as it has less computational complexity. Prediction time is still less; hence it is suitable for developing a model for a decision support system for clinical purposes.

Table 4 Table X shows the training time and prediction time for various conFigureurations of the models. The training time for the stacking ensemble is higher because of the multi-model training process, but the prediction time is minimal

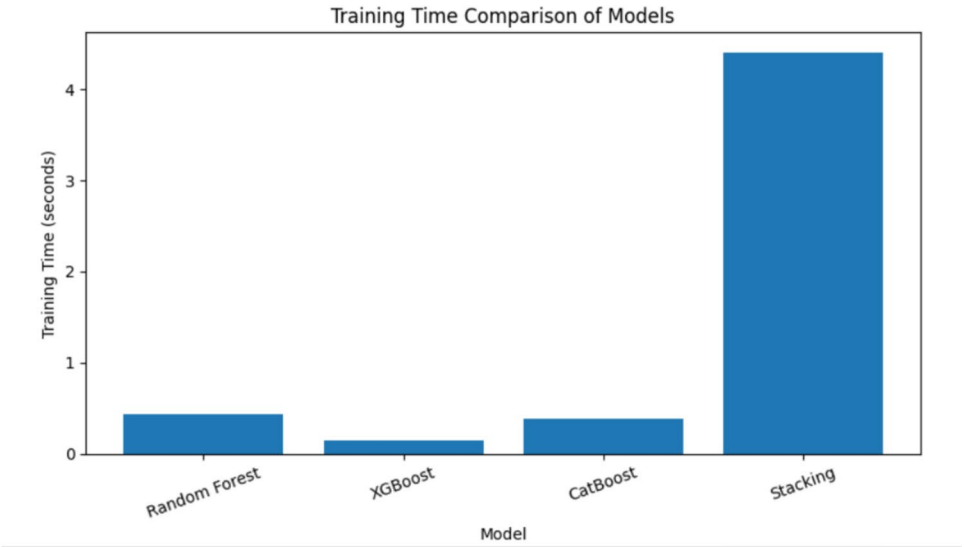


Fig. 11 Training time comparison of different machine learning models

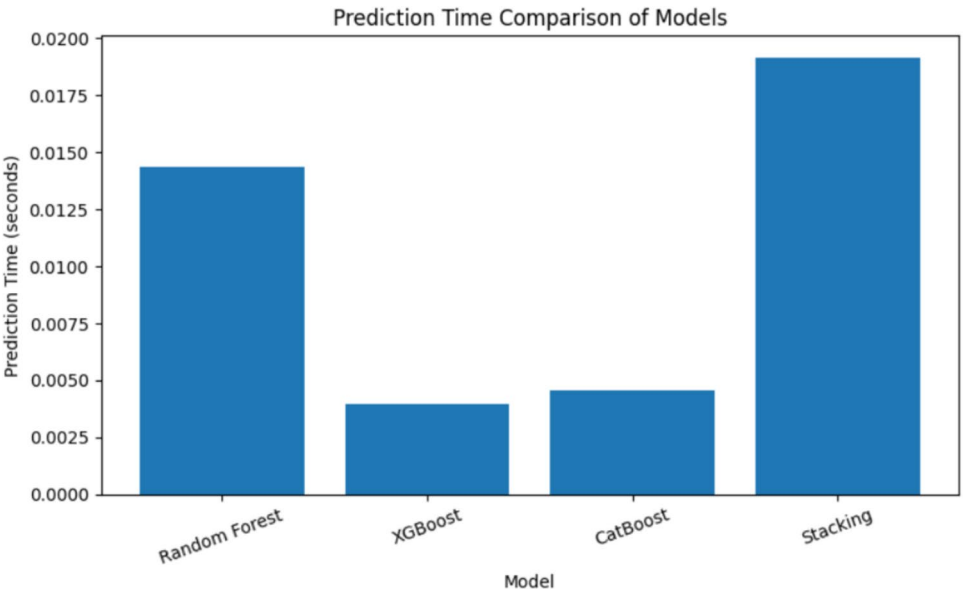


Fig. 12 Predicting time comparison of different machine learning models

From the computational analysis, it is evident that the training time for individual classifiers is lower compared to the stacking ensemble model. Nevertheless, the prediction time for all models is still low, and this indicates that the proposed framework is computationally efficient for implementation. The slight increase in training time is acceptable due to the improvement in the accuracy of the stacking model.

Figures 11 and 12 illustrate the comparison of training/prediction times among various models. The stacking ensemble model has a higher training time because of the learning and meta-level analysis of multiple models. Nevertheless, the prediction time for all models is minimal, which is a positive indication of efficient inference performance. The above findings show that the proposed framework is computationally efficient for real-world implementation.

7 Conclusion

This paper offers an effective stacking ensemble approach for the early prediction of cardiovascular disease based on optimized Machine Learning Models. Even though there is no new Machine Learning Algorithm presented in the paper, the approach shows effectiveness in terms of sensitivity and

robustness. It is clarified that there could be some limitations with regards to the dataset sampling and validation method that would be dealt with in the future.

This study proves that with the right design of the stacking ensemble approach and the usage of meaningful clinical evaluation metrics, the detection of cardiovascular disease at an early stage can be improved without increasing the model's complexity.

7.1 Scope and limitations

Notwithstanding the excellent predictive capability of the proposed stacking ensemble framework, there are some limitations that need to be highlighted. Firstly, the proposed work is conducted on a single benchmark dataset, which may not be very generalizable to other settings. Secondly, the dataset used in the proposed work is restricted to structured clinical variables and does not include other modalities such as imaging or ECG signals, which may further improve the predictive capability of the model.

Furthermore, while the proposed model has excellent predictive capability, the ensemble approach may also make the model less interpretable than a simpler model, which may be a concern in clinical settings. Finally, in order to ensure practical reliability, the proposed model needs to be validated on real-world clinical data.

8 Future work

Future directions will be to test the proposed framework on larger and multi-center cohorts to validate it, using multimodal clinical information such as ECG signals and images, and using cross-validation methods to test its statistical significance. In the classification of cardiovascular diseases, the proposed stacking ensemble framework exhibits promising results; still, a number of possibilities can be explored to increase its efficacy and relevance. Additional clinical parameters, including as electrocardiogram (ECG) readings, patient history, imaging data, and lifestyle factors, can be incorporated into the model in further work to potentially produce more precise and customized predictions. Additionally, feature learning may be enhanced by combining deep learning approaches with conventional ensemble methods, particularly for complicated or unstructured medical data. Using AI techniques like SHAP or LIME to improve model interpretability is another crucial avenue that can help clinicians comprehend and have faith in the model's findings. Finally, in order to cut down dimensionality and increase the computing efficiency of the model without sacrificing accuracy, future research can look into the application of hybrid feature selection techniques.

Author contribution D.M. contributed to the conceptualization of the study, methodology design, data collection, preprocessing, and implementation of machine learning models. A.A. was responsible for feature engineering, hyperparameter optimization, statistical analysis, and preparation of figures and tables. A.M. supervised the research, guided the experimental framework, reviewed and edited the manuscript critically for important intellectual content. All authors reviewed and approved the final version of the manuscript.

Funding There were no funding or grant received for this manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interest The authors declare no competing interests.

References

1. World Health Organization, "Cardiovascular Diseases (CVDs)," [Online]. Available: https://www.afro.who.int/health-topics/cardiovascular-diseases. Accessed: May 5, 2023.
2. Alom Z et al. Early Stage Detection of Heart Failure Using Machine Learning Techniques, in Proc Int Conf Big Data, IoT, Mach Learn, Cox's Bazar, Bangladesh, Sep. 23–25, 2021.
3. Gour S, Panwar P, Dwivedi D, Mali C, et al. A Machine Learning Approach for Heart Attack Prediction. In: Nagar AK, et al., editors. Intelligent Sustainable Systems. Singapore: Springer; 2022. p. 741–7.
4. Gupta C, Saha A, Reddy NS, Acharya UD. Cardiac disease prediction using supervised machine learning techniques. J Phys Conf Ser. 2022;2161:012013.
5. Shameer K, et al. Machine learning predictions of cardiovascular disease risk in a multi-ethnic population using electronic health record data. Int J Med Inform. 2021;146:104335.
6. Ali F, et al. A smart healthcare monitoring system for heart disease prediction based on ensemble deep learning and feature fusion. Inf Fusion. 2020;63:208–22.
7. Ayon SI, Islam MM, Hossain MR. Coronary artery heart disease prediction: a comparative study of computational intelligence techniques. IETE J Res. 2020;68(4):1–20.
8. Rong G, et al. Artificial intelligence in healthcare: review and prediction case studies. Engineering. 2020;6(3):291–301.
9. Fitriyani NL, et al. HDPM: An effective heart disease prediction model for a clinical decision support system. IEEE Access. 2020;8:133034–50.
10. Benjamin et al. EJ. Heart disease and stroke statistics—2019 update: a report from the American Heart Association. Circulation 2019;139(10).
11. Ke C, et al. Divergent trends in ischaemic heart disease and stroke mortality in India from 2000 to 2015: a nationally representative mortality study. Lancet Glob Health. 2018;6(8):e914–23.
12. Bharti R et al. Prediction of heart disease using a combination of machine learning and deep learning. Comput Intell Neurosci. 2021;2021:8387680.

13. Sharifi E, et al. Preparation of a biomimetic composite scaffold from gelatin/collagen and bioactive glass fibers for bone tissue engineering. Mater Sci Eng C Mater Biol Appl. 2016;59:533–41.
14. Ramanathan VA, Yerra S, Swetha NG, Gandikota P. Advancements in heart disease prediction: a machine learning approach for early detection and risk assessment. Intl J Sci Res (IJSR). 2024;13(10):774–80. https://doi.org/10.21275/SR241004171836.
15. Narayanan J. Implementation of efficient machine learning techniques for prediction of cardiac disease using SMOTE. Procedia Comput Sci. 2024;233:558–69. https://doi.org/10.1016/j.procs.2024.03.245.
16. El-Sofany H, Bouallegue B, El-Latif YMA. A proposed technique for predicting heart disease using machine learning algorithms and an explainable AI method. Sci Rep. 2024;14:23277. https://doi.org/10.1038/s41598-024-74656-2.
17. Nandy S, Adhikari M, Balasubramanian V, Menon VG, Li X, Zakarya M. An intelligent heart disease prediction system based on swarm-artificial neural network. Neural Comput Applic. 1 (May 2021). https://doi.org/10.1007/s00521-021-06124-1.
18. Long NC, Meesad P, Unger H. A highly accurate firefly based algorithm for heart disease prediction. Expert Syst Appl. 2015;42(21):8221–31. https://doi.org/10.1016/j.eswa.2015.06.024.
19. Ayon SI, Islam MM, Hossain MR. Coronary artery heart disease prediction: a comparative study of computational intelligence techniques. IETE J Res. 2020;68(4):1–20. https://doi.org/10.1080/03772063.2020.1713916.
20. Srinivas P, Katarya R. hyOPTXg: OPTUNA hyper-parameter optimization framework for predicting cardiovascular disease using XGBoost. Biomed Signal Process Control. 2022;73:103456.
21. Amin MS, Chiam YK, Varathan KD. Identification of significant features and data mining techniques in predicting heart disease. Telemat Inform. 2019;36:82–93.
22. Mohan S, Thirumalai C, Srivastava G. Effective heart disease prediction using hybrid machine learning techniques. IEEE Access. 2019;7:81542–54.
23. Javeed A, et al. An intelligent learning system based on random search algorithm and optimized random forest model for improved heart disease detection. IEEE Access. 2019;7:18073–85.
24. Budholiya K, Shrivastava SK, Sharma V. An optimized XGBoost based diagnostic system for effective prediction of heart disease. J King Saud Univ Comput Inf Sci. 2022;34(7):4514–23.
25. Ali L, et al. An optimized stacked support vector machines based expert system for the effective prediction of heart failure. IEEE Access. 2019;7:54007–14.
26. Ali L, et al. An automated diagnostic system for heart disease prediction based on χ^2 statistical model and optimally conFigureured deep neural network. IEEE Access. 2019;7:34938–45.
27. Dileep P, et al. An automatic heart disease prediction using cluster-based bidirectional LSTM (C-BiLSTM) algorithm. Neural Comput Appl. 2022. https://doi.org/10.1007/s00521-022-07064-0.
28. Azmi J, et al. A systematic review on machine learning approaches for cardiovascular disease prediction using medical big data. Med Eng Phys. 2022;105:103825.
29. Ahmad GN, et al. Efficient medical diagnosis of human heart diseases using machine learning techniques with and without GridSearchCV. IEEE Access. 2022;10:316579–89.
30. Ali MM, et al. Heart disease prediction using supervised machine learning algorithms: performance analysis and comparison. Comput Biol Med. 2021;136:104672.
31. El-Hasnony IM, et al. Multi-label active learning-based machine learning model for heart disease prediction. Sensors. 2022;22(3):1184.
32. El-Shafiey MG, et al. A hybrid GA and PSO optimized approach for heart-disease prediction based on random forest. Multimed Tools Appl. 2022;81(13):18155–79.
33. Deepika D, Balaji N. Effective heart disease prediction using novel MLP-EBMDA approach. Biomed Signal Process Control. 2022;72:103318.
34. Hasanova H, et al. A novel blockchain-enabled heart disease prediction mechanism using machine learning. Comput Electr Eng. 2022;101:108086.
35. Al-Tam RM, Al-Hejri AM, Naji E, Hashim FA, Alshamrani SS, Alshehri A, Narangale SM (2024) A hybrid framework of transformer encoder and residential conventional for cardiovascular disease recognition using heart sounds. IEEE Access.
36. Olawade DB, Soladoye AA, Omodunbi BA, Aderinto N, Adeyanju IA (2025) Comparative analysis of machine learning models for coronary artery disease prediction with optimized feature selection. Intl J Cardiol. 133443.
37. Kumar AS, Rekha R. A dense network approach with Gaussian optimizer for cardiovascular disease prediction. New Gener Comput. 2023;41:859–78.
38. Saran Kumar A, Rekha R. An improved hawks optimizer based learning algorithms for cardiovascular disease prediction. Biomed Signal Process Control. 2023;81:104442.
39. Arunachalam SK, Rekha R. A novel approach for cardiovascular disease prediction using machine learning algorithms. Concurrency Comput Pract Exp. 2022;34(19):e7027.
40. Saranya K, Karthikeyan U, Kumar AS, et al. DenseNet-ABiLSTM: revolutionizing multiclass arrhythmia detection and classification using hybrid deep learning approach leveraging PPG signals. Int J Comput Intell Syst. 2025;18:33. https://doi.org/10.1007/s44196-025-00765-z.
41. Rajagopal R, Ranganathan V. Evaluation of effect of unsupervised dimensionality reduction techniques on automated arrhythmia classification. Biomed Signal Process Control. 2017;34:1–8. https://doi.org/10.1016/j.bspc.2016.12.017.
42. Rani P, et al. A decision support system for heart disease prediction based upon machine learning. J Reliab Intell Environ. 2021;7(3):263–75.
43. Dangare CS, Apte SS. Improved study of heart disease prediction system using data mining classification techniques. Int J Comput Appl. 2012;47(10):44–8.
44. Valarmathi R, Sheela T. Heart disease prediction using hyper parameter optimization (HPO) tuning. Biomed Signal Process Control. 2021;70:103033.
45. Patel J, et al. Heart disease prediction using machine learning. Lecture Notes in Networks and Systems. 2021;203:653–65.
46. Benhar H, Idri A, Fernández-Alemán JL. Data preprocessing for heart disease classification: a systematic literature review. Comput Methods Programs Biomed. 2020;195:105635.
47. Habib U et al. Outliers detection method using clustering in buildings data, in Proc. 41st Annu. Conf. IEEE Ind. Electron. Soc. (IECON). 2015;694–700.
48. Dhananjay B, Sivaraman J. Analysis and classification of heart rate using CatBoost feature ranking model. Biomed Signal Process Control. 2021;68:102610.
49. Chen T, Guestrin C. XGBoost: A scalable tree boosting system, in Proc 22nd ACM SIGKDD Int Conf Knowl Discov Data Min, 2016.
50. Jiang Y, et al. A pedestrian detection method based on genetic algorithm for optimize XGBoost training parameters. IEEE Access. 2019;7:118310–21.
51. Li S, Zhang X. Research on orthopedic auxiliary classification and prediction model based on XGBoost algorithm. Neural Comput Appl. 2020;32(7):1971–9.

52. Liu P, et al. Optimizing survival analysis of XGBoost for ties to predict disease progression of breast cancer. IEEE Trans Biomed Eng. 2021;68(1):148–60.
53. Binson VA, et al. Prediction of pulmonary diseases with electronic nose using SVM and XGBoost. IEEE Sensors J. 2021;21(18):20886–95.
54. Habib AZSB, Tasnim, T Billah MM. A study on coronary disease prediction using boosting-based ensemble machine learning approaches, in Proc 2nd Int Conf Innovation Eng Technol (ICIET), Dhaka, Bangladesh, 2019;1–6.
55. Zhang D, Gong Y. The comparison of LightGBM and XGBoost coupling factor analysis and prediagnosis of acute liver failure. IEEE Access. 2020. https://doi.org/10.1109/access.2020.3042848.
56. Zhang S, et al. Improvement of the performance of models for predicting coronary artery disease based on XGBoost algorithm and feature processing technology. Electronics. 2022. https://doi.org/10.3390/electronics11030315.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.